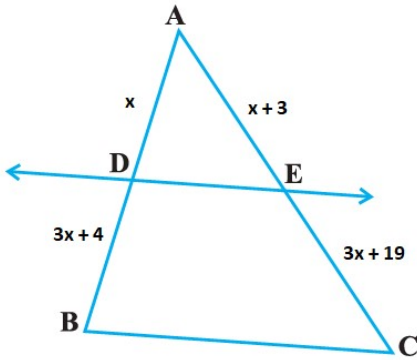
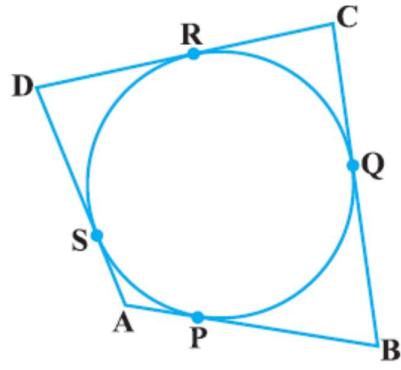
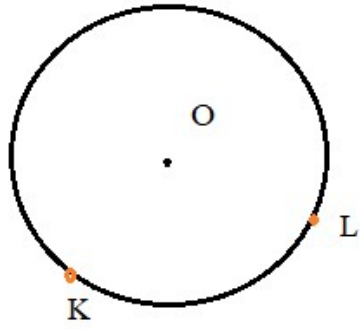


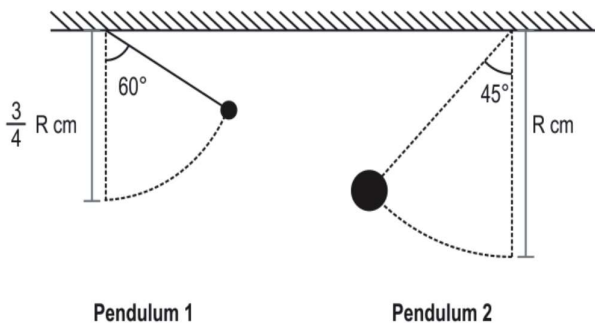
Pre – Board Examination (2024-25)**Class- X****Time: 3 hours****Subject- Maths(Standard)****MM - 80****General Instructions:****Read the following instructions carefully and follow them:**

1. This question paper contains 38 questions.
2. This Question Paper is divided into 5 Sections A, B, C, D and E.
3. In Section A, Questions no. 1-18 are multiple choice questions (MCQs) and questions no. 19 and 20 are Assertion- Reason based questions of 1 mark each.
4. In Section B, Questions no. 21-25 are very short answer (VSA) type questions, carrying 02 marks each.
5. In Section C, Questions no. 26-31 are short answer (SA) type questions, carrying 03 marks each.
6. In Section D, Questions no. 32-35 are long answer (LA) type questions, carrying 05 marks each.
7. In Section E, Questions no. 36-38 are case study based questions carrying 4 marks each with sub parts of the values of 1, 1 and 2 marks each respectively.
8. All Questions are compulsory. However, an internal choice in 2 Question of Section B, 2 Questions of Section C and 2 Questions of Section D has been provided. An internal choice has been provided in all the 2 marks questions of Section E.
9. Draw neat and clean figures wherever required.
10. Take $\pi = 22/7$ wherever required if not stated.
11. Use of calculators is not allowed.

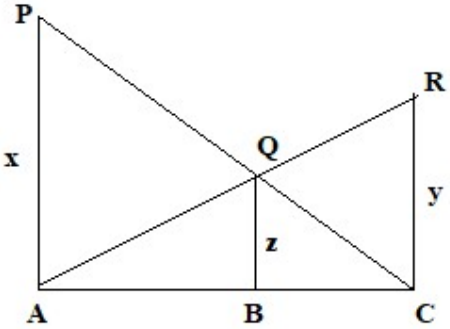
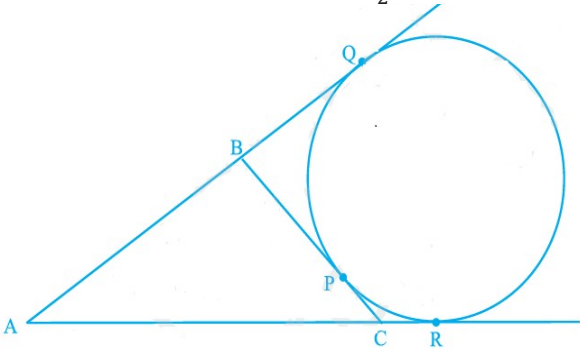
SECTION - A**Section A consists of 20 questions of 1 mark each.**

S.NO		Marks
1	The HCF of two numbers 65 and 104 is 13. If LCM of 65 and 104 is $40x$, then the value of 'x' is (A) 5 (B) 13 (C) 40 (D) 8	1
2	For what value of ' k ' will $k+9$, $2k-1$ and $2k+7$ are the consecutive terms of an AP ? (A) 18 (B) 13 (C) 8 (D) 10	1
3	The two zeroes of the polynomial $2x^2 - 6x - 3$ are of the form $\frac{3 \pm \sqrt{k}}{2}$, the value of 'k' is (A) 5 (B) 15 (C) -15 (D) 8	1
4	The roots of the quadratic equation $y^2 + y - 1 = 0$ are (A) Irrational and distinct (B) not real (C) rational and distinct (D) real and equal	1
5	A card is drawn from a well shuffled deck of cards. What is the probability that the card drawn is neither a king nor a queen? (A) $\frac{11}{13}$ (B) $\frac{12}{13}$ (C) $\frac{11}{26}$ (D) $\frac{11}{52}$	1
6	Find the value of x for which $DE \parallel BC$ in the given figure. (A) 1 (B) 2 (C) 3 (D) 4	1

		
7	A quadratic polynomial whose zeroes are $3 + \sqrt{2}$ and $3 - \sqrt{2}$ is (A) $x^2 - 6x + 7$ (B) $x^2 + 6x + 7$ (C) $x^2 - 6x - 7$ (D) $x^2 + 6x - 7$	1
8	If $\sin(A - B) = \frac{1}{2}$ and $\cos(A + B) = \frac{1}{2}$, $0^\circ < A + B < 90^\circ$, $A > B$, then A and B are respectively. (A) $30^\circ, 45^\circ$ (B) $45^\circ, 30^\circ$ (C) $45^\circ, 15^\circ$ (D) $60^\circ, 15^\circ$	1
9	In the given figure if AB = 6 cm, BC = 7 cm and CD = 4 cm, then length of AD is  (A) 3 cm (B) 4 cm (C) 5 cm (D) 6 cm	1
10	If $P(\frac{a}{3}, 4)$ is the mid-point of the line segment joining the points $Q(-6, 5)$ and $R(-2, 3)$, then the value of 'a' is (A) -4 (B) -12 (C) 12 (D) -6	1
11	If the system of equations $3x + y = 1$ and $(2k - 1)x + (k - 1)y = 2k + 1$ is inconsistent, then k = (A) -1 (B) 0 (C) 1 (D) 2	1
12	A circle is shown below:-  (i) There is only one line passing through point K which makes an angle of 90° with OK. (ii) The shortest distance of a tangent passing through point L from the centre O is equal to the radius of the circle, OL. (iii) One tangent can pass through two points K and L of a circle. Which statement(s) is /are correct. (A) (i) and (ii) (B) (iii) (C) (ii) and (iii) (D) (i), (ii), (iii)	1

13	The value of $2\sin^2 30^\circ - 3\cos^2 45^\circ + \tan^2 60^\circ + 3\sin^2 90^\circ$ is (A) 1 (B) 5 (C) 0 (D) none of these	1												
14	Shown below are two pendulums of different length attached to a bar  Pendulum 1 Pendulum 2 Based on the figure shown above, the arc length of pendulum 1 is _____ the arc length of pendulum 2. (A) greater than (B) lesser than (C) equal to (D) can not be answered without knowing the value of R	1												
15	The volume of the largest sphere that can be carved out of a cube of side 7cm is (A) 179.66 cm^3 (B) 279.66 cm^3 (C) 169.66 cm^3 (D) 205 cm^3	1												
16	For the following distribution: <table border="1" style="width: 100%; border-collapse: collapse; text-align: center;"><tr><td>Class</td><td>0-5</td><td>5-10</td><td>10-15</td><td>15-20</td><td>20-25</td></tr><tr><td>Frequency</td><td>10</td><td>15</td><td>12</td><td>20</td><td>9</td></tr></table> The sum of lower limits of the median class and modal class is (A) 15 (B) 25 (C) 30 (D) 35	Class	0-5	5-10	10-15	15-20	20-25	Frequency	10	15	12	20	9	1
Class	0-5	5-10	10-15	15-20	20-25									
Frequency	10	15	12	20	9									
17	The mean of five numbers is 15. If we include one more number, the mean of six numbers becomes 17. The included number is (A) 27 (B) 37 (C) 17 (D) 25	1												
18	A bag contains 100 cards numbered 1 to 100. A card is drawn at random from the bag. What is the probability that the number on the card is a perfect cube? (A) $\frac{1}{20}$ (B) $\frac{3}{50}$ (C) $\frac{1}{25}$ (D) $\frac{7}{100}$	1												
	DIRECTION: In the question number 19 and 20, a statement of Assertion (A) is followed by a statement of Reason (R). Choose the correct option (A) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A) (B) Both assertion (A) and reason (R) are true and reason (R) is not the correct explanation of assertion (A) (C) Assertion (A) is true but reason (R) is false. (D) Assertion (A) is false but reason (R) is true.													
19	Assertion (A) : If the points A(4,3) and B(x,5) lie on a circle with centre O(2,3), then the value of 'x' is 2 Reason (R) : Centre of a circle is the mid- point of each chord of the circle.	1												
20	Assertion (A) : If the volumes of two spheres are in the ratio 27:8. Then their surface areas are in the ratio 3:2. Reason (R) : Volume of the sphere $= \frac{4}{3}\pi r^3$ and its surface area $= 4\pi r^2$	1												
SECTION B														
Section B consists of 5 questions of 2 marks each.														
S.No		Marks												
21	Find the HCF of 612 and 1314 using prime factorisation method. Or Explain whether $3 \times 12 \times 101 + 4$ is a prime number or composite number.	2												
22	Show that the points A (5,1), B(1,4) and C(8,5) are the vertices of a right- angled triangle.	2												

23	Find the ratio in which the segment joining the points (1,-3) and (4,5) is divided by x-axis? Also find the coordinates of this point on x – axis.	2
24	If $\sin\theta + \cos\theta = \sqrt{2}$, then find $\tan\theta + \cot\theta$.	2
25	Tree Plantation Drive A Group Housing society has 600 members, who have their houses in the campus and decided to hold a tree plantation drive on the occasion of new year. Each household was given the choice of planting a sampling of its choice. The number of different types of samplings planted were: (i) Neem = 125 (ii) Peepal = 165 (iii) Creepers = 50 (iv) Fruit plants = 150 (v) Flowering plants = 110 On the opening ceremony, one of the plants is selected randomly for a prize. What is the probability that the selected plant is (A) a fruit plant or a flowering plant? (B) either a Neem plant or a Peepal plant? Or Three unbiased coins are tossed together. Find the probability of getting: (A) two heads (B) at most two tails	2
SECTION C		
Section C consists of 6 questions of 3 marks each.		
S.No		Marks
26	An arc of a circle of radius 10 cm subtends an angle of 60° at the centre. Find (i) the length of the arc (ii) the area of the minor segment of the circle made by the corresponding chord. Or Rahul bought a pair of glasses with wiper blades. He was curious to know the area being cleaned by each of wiper blades. With the help of ruler and a protractor, he found the length of each blade as 3 cm and the angle swept as 60°.  (i) Find the area that each wiper cleans in one swipe, in terms of π. (ii) If the diameter of each circular glass is 5 cm, what per cent(%) of the area of the glass will be cleaned by the blade in one swipe?	3
27	Prove that $\sqrt{2} + \sqrt{3}$ is irrational.	3
28	The sum of two numbers is 9. The sum of their reciprocals is $\frac{1}{2}$. Find the numbers.	3
29	In given figure, PA, QB and RC are each perpendicular to AC. If PA = x units, RC = y units and QB = z units. Prove that: $\frac{1}{x} + \frac{1}{y} = \frac{1}{z}$.	

	 Or Sides AB and BC and median AD of a triangle ABC are respectively proportional to sides PQ and QR and median PM of triangle PQR. Show that $\triangle ABC \sim \triangle PQR$.	3
30	Quadratic polynomial $2x^2 - 3x + 1$ has zeros as α and β . Now form a quadratic polynomial whose zeros are 3α and 3β .	3
31	Prove that : $\frac{\sin A + \cos A}{\sin A - \cos A} + \frac{\sin A - \cos A}{\sin A + \cos A} = \frac{2}{2\sin^2 A - 1}$	3
SECTION D		
Section D consists of 4 questions of 5 marks each.		
S.No		Marks
32	Prove that “<i>The lengths of tangents drawn from an external point to a circle are equal.</i>” Hence using the above theorem, prove that If a circle touches the side BC of a triangle ABC at P and extended sides AB and AC at Q and R, respectively, then $AQ = \frac{1}{2}(BC + CA + AB)$ 	2+3
33	Draw the graphs of the pair of linear equations $x - y + 2 = 0$ and $4x - y - 4 = 0$. Calculate the area of the triangle formed by the lines so drawn and the x-axis. Or Kamlesh sold a table and a chair for Rs 1050, thereby making a profit of 10% on the table and 25% on the chair. If he had taken a profit of 25% on the table and 10% on the chair he would have got Rs 1065. Find the cost price of each.	5
34	The angles of depression of the top and the bottom of an 8 m tall building from the top of a multi-storeyed building are 30° and 45° , respectively. Find the height of the multi-storeyed building and the distance between the two buildings.	5
35	A survey regarding the heights (in cm) of 51 girls of Class X of a school was conducted and the following data was obtained. Find the median height.	

Height in cm	Less than 140	Less than 145	Less than 150	Less than 155	Less than 160	Less than 165
Number of girls	4	11	29	40	46	51

OR

The following table shows the ages of the patients admitted in a hospital during a year:

Age (in years)	5-15	15-25	25-35	35-45	45-55	55-65
Number of patients	6	11	21	23	14	5

Find the mode and the mean of the data given above. Hence find the median using empirical relationship.

5

SECTION E

Case study based questions are compulsory.

36

Kumbh Mela is a major pilgrimage and festival in Hinduism. It is celebrated in a cycle of approximately 12 years at four river-bank pilgrimage sites: the Prayagraj (Ganges-Yamuna Sarasvati rivers confluence), Haridwar (Ganges), Nashik (Godavari), and Ujjain (Shipra). The festival is marked by a ritual dip in the waters. The seekers believe that bathing in these rivers is a means to prayascitta for past mistakes, and that it cleanses them of their sins. Government of Uttarakhand is planning to procure tent for the pilgrims during Kumbh Mela.

The specification of tent is- it is in the shape of cylinder surmounted by a conical top. The height and diameter of cylindrical part are 9m and 30m respectively and height of the conical part is 8m with same diameter as that of cylindrical part.

Based on the above information answer the following:

- What is the slant height of conical part?
- Find the area of the canvas used in making the tent.
- Find the cost of canvas bought for the tent at the rate Rs 200 per sq m, if 30 sq m canvas was wasted during stitching.

Or

Find the volume of air in the tent.



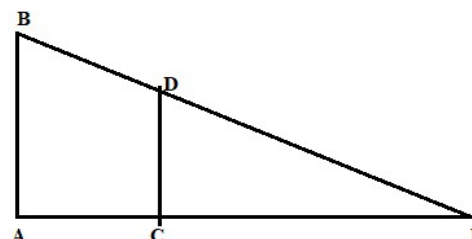
1

1

2

37

Two trees are standing parallel to each other. The bigger tree is 8m high, casts a shadow of 6m.



- If AB and CD are the two trees and AE is the shadow of the longer tree,

	then state whether $\triangle AEB \sim \triangle CED$, give reason. (ii) If the ratio of the height of two trees is 3:1, then find the length of the shadow of the smaller tree. (iii) Find the distance of point B from E. Or If the ratio of the height of two trees 4:1, then find the length of shadow of the smaller tree.	1 1 2
38	India is a competitive manufacturing location due to the low cost of manpower and strong technical and engineering capabilities contributing to higher quality production runs. The production of TV sets in a factory increases uniformly by a fixed number every year. It produced 16000 sets in the 6th year and 22600 in 9th year. Based on the above information answer the following: (i) What is the production during first year? (ii) Find the total production in(during) first 3 years. (iii) In which year, the production was 29200? Or What was the difference in the production during 7th year and the 4th year?	1 1 2